MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6189W 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, medium Mu.

Outline - - - 6-2
Base - - - E9-1
Envelope - - -T6-1/2
Base connections:

Pin No. - - -	1	2	3	4	5	6	7	8	9
Element - - -	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rg/g	Ik/k	Ic/g	PP/p	TE	Alt
Unit:	V	Vdc	Vdc	v	Meg	mAdc	mAdc	W	°C	ft
Maximum:	13.8 6.9	300	0, -50	±100	0.5 (See note 1)	20	5.0	2.7	165	(See note 2)
Minimum	11.4 5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	12.6	250	-8.5	0	---	---	---	---	---	---

Pulse Service.

Parameter:	Ef	Ebb	egy	Pg/g	Du	ik/k	tp	prr	ehk	Ec	Rk/k	Pp/p
Unit:	V	Vdc	V	W	%	ma	µs	---	v	Vdc	Ohm	W
Maximum:	13.8 6.9	300	+60	0.4					100	0,-55	---	3.0
Minimum	11.4 5.7	---	---	---		See figure 1			---	---	---	---
TEST CONDITIONS:	12.6	250	+55	---	1.0	---	10	1,000	0	-45	1.0	---

GENERAL:

Qualification - Required
Reliable tube

1/ Formerly tube type 6189

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)	3,11		0.4	II	Ib	6.5	14.5	mAdc
1256	Electrode current (2) (anode)	4	Ec = -30 Vdc; Rp = 0.1 Meg	0.4	II	Ib	---	20	µAdc
1266	Total grid current	3,11	Rg/g = 0.5 Meg	0.4	II	Ic	0	-0.5	µAdc
1301	Heater current			0.4	II	If	138	162	mA
1306	Transconductance (1)	3		0.4	II	Sm	1,750	2,650	µmhos
1336	Heater-cathode leakage	3		0.4	II	Ihk	---	7	µAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	3		2.5	I	R	1,000	---	Meg
2201	Noise and microphonics	5	Ef = 12.6 Vac; Ebb = 300 Vdc; Ec = 0; Ecal = 50 mVac; Rk = 1,500 ohms; Rl = 50,000 ohms; Rg = 0	2.5	I	---	---	---	---
1256	Electrode current (3) (anode)	3	Ec = -18 Vdc	2.5	I	Ib	5	---	µAdc
1256	Electrode current (1) (anode) (difference between sections)			2.5	I	Ib	---	3.5	mAdc
1266	Grid emission	3,6	Ef = 15.0 V; Ec = -30 Vdc; Rg/g = 0.5 Meg	2.5	I	Isg	0	-1.5	µAdc
1306	Transconductance (2)	3	Ef = 11.4 V	2.5	I	ΔSm Ef	---	10	%
1316	Amplification factor	3		6.5	S3	Mu	15.5	18.5	---
---	Pulse cathode current (1)	3,7		2.5	I	ik Δik tp	400 ---	--- 15	mA %
---	Pulse cathode current (2)	3,7	Ef = 11.4 V	6.5	Code E	ik Δik tp	350 ---	--- 20	ma %
1105	Permanence of marking			---	---	---	---	---	---
1031	High-frequency vibration	5	Rp = 2,000 ohms	6.5	Code E	Ep	---	100	mVac
1031	Sweep-frequency vibration	5	Rp = 2,000 ohms	6.5	See note 10	Ep	---	300	mVac

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 - Continued								
1306	Transconductance (3)	3	Eb = 100 Vdc; Ec = 0	2.5	I	Sm	2,500	4,000	µmhos
1331	Direct-interelectrode capacitance	3	No shield	6.5	Code E	Cgp	1.20	1.80	pF
		3	No shield			Cin	1.25	1.95	pF
			No shield			Cout 1	0.30	0.70	pF
			No shield			Cout 2	0.20	0.60	pF
1121	Base strain	12		---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1041	Shock	8	450 G; Ehk = +100 Vdc	6.5	See note 10	---	---	---	---
1031	Vibration-fatigue			6.5	See note 10	---	---	---	---
---	Shock and vibration-fatigue test end points:								
1031	High-frequency vibration			---	---	Ep	---	150	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	10	µAdc
1306	Transconductance (1)			---	---	Sm	1,600	---	µmhos
1266	Total grid current			---	---	Ic	0	-1.5	µAdc

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; heaters in parallel; Ehk = 135 Vdc; Ec = Eb = 0; 1 min "on", 4 min "off"	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	15	µAdc
1516	Stability life		Ehk = +135 Vdc; Rg/g = 0.5 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSm_t	---	10	%
1501	Intermittent life	13	Ehk = +135 Vdc; Rg/g = 0.5 Meg; TA = room; TE = 165°C (min); Group E	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 -Continued						
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current			Ic	0	-0.5	µAdc
1301	Heater current			If	138	166	mA
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	20	%
1306	Transconductance (2)			ΔSmEf	---	20	%
1336	Heater-cathode leakage			Ihk	---	7	µAdc
1211	Insulation of electrodes			R	250	---	Meg
1306	Transconductance (1) average change			AvgΔSmt	---	10	%
---	Life test (cutoff)	13	Ef =f 12.6 V; Eb = 250 Vdc; Ec = -30 Vdc	---	---	---	---
---	Life test (cutoff) end-points (1,000 hours):						
1266	Total grid current			Ic	0	-0.5	µAdc
1211	Insulation of electrodes			R	250	---	Meg
1501	Intermittent life (grid pulse)	9	Ef = 12.6 V; Eb = 300 Vdc; egk = +60 ±1 V; Rp/p = 180 ohms; Rk = 0; prr = 1,000 pps; TA = room; Group E	---	---	---	---
---	Intermittent life (grid pulse) - test end points (500 hours):						
---	Pulse cathode current (1)			ik	300	---	mA
1266	Total grid current			Ic	0	-0.5	µAdc
1211	Insulation of electrodes			R	500	---	Meg
1511	Cathode interface life	3	Ef = 6.9 V; heaters in parallel; Ebb = 50 V; adjust Ec until Ia = 1.0 mA	Ri	---	50	Ohms

NOTES:

1. This value is for operation under fixed-bias conditions. With cathode bias, Rg/g may be 1.0 megohm maximum.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
3. Test each unit separately.
4. Test each section separately with normal test condition voltages applied to the opposite section.
5. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
6. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ec Vdc	Eb Vdc	Rk/k Ohm	Rg/g Meg
15.0	-8.5	250	0	0.5

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